

APPLIED MACHINE LEARNING

- Python
- XGBoost
- Autoencoders
- Anomaly Detection
- Classification
- Time Series
- Forecasting
- Computer Vision

LLM SYSTEMS

- Transformers
- LoRA / QLoRA
- PEFT
- Unsloth
- vLLM
- LLM Evaluation
- Information Extraction
- OpenAI-Compatible APIs

PRODUCTION ML

- GCP
- GKE
- Vertex AI
- Kubernetes
- Seldon Core
- MLflow
- Dataflow / Beam

EVALUATION & RELIABILITY

- Model Monitoring
- Prometheus
- Grafana
- CI/CD
- GitOps
- ArgoCD
- Istio
- Blue-Green Delivery

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024
Generative AI Certified Professional
- Microsoft Certified
Azure AI Engineer Associate
- Microsoft Certified
Azure Data Scientist Associate
- Google Cloud
Certified Generative AI Leader

EDUCATION

- B.E. Electrical & Electronics
Andhra University | 2012 - 2016

ML METHODS

- Scikit-learn
- PyTorch
- TensorFlow
- Keras
- DenseNet
- Predictive Modelling
- Unsupervised Learning
- Feature Engineering
- Transfer Learning
- NLP
- Time-Series Forecasting

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

EXPERIENCE

Lead Artificial Intelligence Engineer

Best Buy India

Jul 2025 - Present | Bengaluru

- Built a config-driven framework for LoRA/QLoRA fine-tuning across classification, information extraction, and generation tasks using Unsloth, PEFT, Transformers, and Accelerate.
- Automated post-training evaluation with accuracy, confusion-matrix, and MSE outputs; standardized artifacts for hot-loading into a shared vLLM OpenAI-compatible service, with estimated annual API-cost savings of **~\$300K**. Kept model inputs company-managed and reduced external API exposure during peak demand.
- Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
- Owned model rollout, HPA scaling, Istio routing, blue-green delivery, Cloud Build, ArgoCD, Prometheus, Grafana, and production load testing.

Lead ML Engineer — Machine Learning & MLOps

Wipro

Aug 2023 - Jul 2025 | Hyderabad

- Built continuously retrained behavioral autoencoders on NVIDIA DGX (8xA100), processing **billion-row** AWS CloudTrail datasets with Morpheus, RAPIDS, MLflow, and Grafana monitoring.
- Delivered phishing classification at **2,400 emails/sec** - 2x prior throughput - with feedback-aware training and inference pipelines integrated with Microsoft Sentinel.
- Engineered multiformat abuse-inbox classification and information extraction across text, HTML, images, PDF, PNG, and JPEG using Python, OCR, and GPU-accelerated processing.
- Led ML/MLOps delivery and mentoring at Wipro; completed **300+ résumé screens** and **50+ technical interviews**.

Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 - Sep 2023 | India

- Led a production quality-prediction system with automated retraining, monitoring, and stakeholder delivery; increased inspections from **130 to 160 trucks/day**, improved low-quality detection 17%, and generated **\$2.5M annual savings**.
- Built applied ML and optimization systems using Python, Azure ML, Azure Functions, Logic Apps, CI/CD, and heterogeneous operational data.

PROJECTS

LLM Training, Evaluation & Serving

Best Buy | LoRA | vLLM | GKE

A shared path from task configuration and fine-tuning through evaluation, artifact handling, autoscaled serving, controlled rollout, and production observability.

Behavioral Anomaly Detection

Wipro | DGX | 1B+ rows

Per-entity autoencoder models, continuously retrained on high-cardinality CloudTrail data and monitored through operational security dashboards.

GPU STACK

- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Multi-GPU Training

OPTIMISATION & SIMULATION

- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Constraint Modelling
- Operations Research
- Assignment Models
- Supply-Chain Optimization
- Scenario Analysis

CLOUD & MANAGED ML

- GCS
- Artifact Registry
- Azure ML
- Azure DevOps
- Function Apps
- Logic Apps
- Cloud Build
- Google KCC

DATA & ML LIFECYCLE

- Apache Beam
- Dataflow
- Pub/Sub
- Vertex AI Feature Store
- Streaming Pipelines
- Training Pipelines
- Automated Evaluation
- Experiment Tracking
- Model Registry
- Model Versioning
- Continuous Training

LANGUAGES & DATA

- SQL
- Pandas
- NumPy
- MS SQL Server
- IBM DB2
- Power BI

ENGINEERING LEADERSHIP

- Architecture Reviews
- Technical Mentoring
- Technical Hiring
- Team Building
- Stakeholder Alignment
- Cross-Functional Delivery

ACHIEVEMENTS

15 honors and awards across Best Buy, Wipro, Accenture, and Infosys.
Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.
Best Buy client AML Hall of Fame (2024): DGX delivery during the Wipro engagement.

ADDITIONAL PROJECTS & EXPERIENCE

Document Classification & Extraction
Wipro | OCR | Python

Classification, OCR, normalization, and identifier extraction across noisy email bodies and attachments, routed into real security-response workflows.

Forecasting & Quality ML
GlobalData / Accenture | Production ML

Domain-aware forecasting across 15+ verticals and monitored quality prediction with automated retraining and measurable operational outcomes.

EARLIER EXPERIENCE & ACHIEVEMENTS

Data Scientist
GD Research Centre | May 2020 - May 2021

Built domain-aware time-series forecasting and scenario analysis across energy, manufacturing, healthcare, finance, retail, telecommunications, automotive, and other industry intelligence centers.

Deep Learning Engineer
Tata Consultancy Services | Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML
Infosys | May 2016 - Jul 2019

Built an ML-powered lead-conversion prediction system and business-intelligence applications that moved sales prioritisation from intuition to data.

Achievements
15 honors and awards

Best Buy Talent Quotient (2026); Best Buy client AML Hall of Fame (2024), during the Wipro engagement.

Vehicle Routing & Transport Planning
Accenture | Google OR-Tools | Operations research

Designed vehicle-routing and mixed-integer models covering capacity, time windows, waiting time, demand, and departure constraints, then automated daily operational planning.

Industry Forecasting Platform
GD Research Centre | Time series | Scenario analysis

Designed and deployed domain-aware forecasting across 15+ industries, combining feature engineering, time-series modelling, and scenario analysis for enterprise decisions.

System Ownership
Design | Implementation | Production operations | Maintenance

Owned AI/ML systems from data and training pipelines through evaluation, deployment, monitoring, maintenance, incident support, and continuous improvement.

Engineering Leadership & Team Building
Architecture | Mentoring | Hiring | Cross-functional delivery

Stayed hands-on while leading ML/MLOps delivery, mentoring engineers, coordinating client and platform stakeholders, and helping scale the India AI engineering hub during the Wipro engagement through **300+ résumé screens** and **50+ technical interviews**.